# Fall 2025 EE582 Homework 02 Report
# Nodal-Analysis-based Modeling of Electric Circuits

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Due: Tuesday, October 7, 2025

## Contents

# Section A (Numerical Oscillations of Discretization Rules): Question 1

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at $t = 0$ and the voltage source takes its value at $t = \Delta t$.

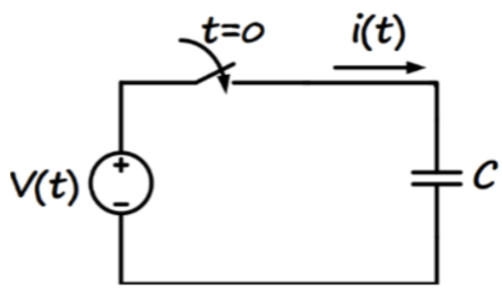


Figure 1: Simple capacitor circuit with DC voltage source

1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance $C$, time-step size $\Delta t$, and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.

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**Note**: I'm assuming that the capacitor has an initial voltage $V_0$ at $t = 0^-$ for generating the state variable trajectory tables. However in question 2, I simulate the case with $V_0 = 0$ V.

### A.1.1 Trapezoidal Discretization of Capacitor

The equations for the Trapezoidal discretization of a capacitor are (see Appendix I for derivation):

**Trapezoidal: Capacitor Companion Model**

$$\boxed{v(t) = e_h + R_C i(t)} \tag{1}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{2}$$

where:

$$R_C = \frac{\Delta t}{2C} \quad \text{(equivalent resistance)} \tag{3}$$

$$e_h = v(t-\Delta t) + R_C i(t-\Delta t) \quad \text{(history voltage)} \tag{4}$$

$$i_h = \frac{v(t-\Delta t)}{R_C} - i(t-\Delta t) \quad \text{(history current)} \tag{5}$$

### Step-by-Step Solution Table for Trapezoidal Discretization

Table 1 shows the step-by-step evolution of the system variables based on trapezoidal discretization.

| $t$ | $v(t)$ | $i(t)$ | $e_h$ | $i_h$ |
|---|---|---|---|---|
| -$h$ | $V_0$ | 0 | $-$ | $-$ |
| $0^+$ | $V_{DC}$ | $\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$ | $V_0$ | $\frac{V_0}{R_C}$ |
| $h$ | $V_{DC}$ | $-\frac{V_{DC}}{R_C} + \frac{V_0}{R_C}$ | $2V_{DC} - V_0$ | $\frac{2V_{DC}}{R_C} - \frac{V_0}{R_C}$ |
| $2h$ | $V_{DC}$ | $\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$ | $V_0$ | $\frac{V_0}{R_C}$ |
| $3h$ | $V_{DC}$ | $-\frac{V_{DC}}{R_C} + \frac{V_0}{R_C}$ | $2V_{DC} - V_0$ | $\frac{2V_{DC}}{R_C} - \frac{V_0}{R_C}$ |
| $4h$ | $V_{DC}$ | $\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$ | $V_0$ | $\frac{V_0}{R_C}$ |

Table 1: Step-by-step solution for trapezoidal discretization of circuit in Figure 1

### A.1.2 Backward-Euler (BE) Discretization of Capacitor

The equations for the Backward-Euler discretization of a capacitor are (see Appendix II for derivation):

**Backward Euler: Capacitor Companion Model**

$$\boxed{v(t) = e_h + R_C\, i(t)} \tag{6}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{7}$$

where:

$$R_C = \frac{\Delta t}{C} \tag{8}$$

$$e_h = v(t - \Delta t) \tag{9}$$

$$i_h = \frac{v(t - \Delta t)}{R_C} \tag{10}$$

#### Step-by-Step Solution Table for Backward-Euler Discretization

Table 2 shows the step-by-step evolution of the system variables based on backward euler discretization.

### A.1.3 Forward Euler (FE) Discretization of Capacitor

Upon solving for the forward euler discretization of the capacitor, we encounter the issue of division by zero as $v(t)$ and $e_h(t)$ are equal at every time step (one source connected to another source directly), and $i(t)$ becomes undefined (because of lack of any resistance between them). $i(t)$ could be of an arbitrarily high magnitude, however in order to ascertain its sign, we introduce a very small resistance $R_\epsilon \ll R_c$ in series with the capacitor to avoid division by zero. This resistance is artificial and does not represent any physical element in the circuit (or at best could be argued to represent the relatively negligible resistance of the connecting wires).

The equations for the Forward Euler discretization of a capacitor are (see Appendix III for derivation):

| $t$ | $v(t)$ | $i(t)$ | $e_h$ | $i_h$ |
|---|---|---|---|---|
| $-h$ | $V_0$ | $0$ | $-$ | $-$ |
| $0^+$ | $V_{DC}$ | $\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$ | $V_0$ | $\frac{V_0}{R_C}$ |
| $h$ | $V_{DC}$ | $0$ | $V_{DC}$ | $\frac{V_0}{R_C}$ |
| $2h$ | $V_{DC}$ | $0$ | $V_{DC}$ | $\frac{V_{DC}}{R_C}$ |
| $3h$ | $V_{DC}$ | $0$ | $V_{DC}$ | $\frac{V_{DC}}{R_C}$ |
| $4h$ | $V_{DC}$ | $0$ | $V_{DC}$ | $\frac{V_{DC}}{R_C}$ |

Table 2: Step-by-step solution for backward euler discretization of circuit in Figure 1

**Forward Euler: Capacitor Companion Model**

$$\boxed{v(t) = e_h(t)} \tag{11}$$

$$\boxed{i(t) = \frac{v(t)}{R_\epsilon} - i_h(t)} \tag{12}$$

where:

$$R_c = \frac{h}{C} \quad \text{(discretized capacitor resistance)} \tag{13}$$

$$R_\epsilon \ll \frac{h}{C} \quad \text{(artificial resistance)} \tag{14}$$

$$e_h(t) = v(t-h) + R_c i(t-h) \tag{15}$$

$$i_h(t) = \frac{e_h(t)}{R_\epsilon} \tag{16}$$

### Step-by-Step Solution Table for Forward-Euler Discretization

Table 3 shows the step-by-step evolution of the system variables based on forward euler discretization, assuming a small resistance $R_\epsilon$ to avoid division by zero.

| $t$ | $v(t)$ | $i(t)$ | $e_h(t)$ | $i_h(t)$ |
|---|---|---|---|---|
| $-h$ | $V_0$ | $0$ | $-$ | $-$ |
| $0^+$ | $V_{DC}$ | $\dfrac{V_{DC}-V_0}{R_\epsilon}$ | $V_0$ | $\dfrac{V_0}{R_\epsilon}$ |
| $h$ | $V_{DC}$ | $\dfrac{V_0-V_{DC}}{R_\epsilon^2/R_c}$ | $V_{DC}\left(1+\frac{R_c}{R_\epsilon}\right)-V_0\left(\frac{R_c}{R_\epsilon}\right)$ | $\dfrac{V_{DC}\left(1+\frac{R_c}{R_\epsilon}\right)-V_0\left(\frac{R_c}{R_\epsilon}\right)}{R_\epsilon}$ |
| $2h$ | $V_{DC}$ | $\dfrac{V_{DC}-V_0}{R_\epsilon^3/R_c^2}$ | $-V_{DC}\left(\frac{R_c^2}{R_\epsilon^2}-1\right)+V_0\left(\frac{R_c^2}{R_\epsilon^2}\right)$ | $\dfrac{-V_{DC}\left(\frac{R_c^2}{R_\epsilon^2}-1\right)+V_0\left(\frac{R_c^2}{R_\epsilon^2}\right)}{R_\epsilon}$ |
| $3h$ | $V_{DC}$ | $\dfrac{V_0-V_{DC}}{R_\epsilon^4/R_c^3}$ | $V_{DC}\left(1+\frac{R_c^3}{R_\epsilon^3}\right)-V_0\left(\frac{R_c^3}{R_\epsilon^3}\right)$ | $\dfrac{V_{DC}\left(1+\frac{R_c^3}{R_\epsilon^3}\right)-V_0\left(\frac{R_c^3}{R_\epsilon^3}\right)}{R_\epsilon}$ |
| $4h$ | $V_{DC}$ | $\dfrac{V_{DC}-V_0}{R_\epsilon^5/R_c^4}$ | $-V_{DC}\left(\frac{R_c^4}{R_\epsilon^4}-1\right)+V_0\left(\frac{R_c^4}{R_\epsilon^4}\right)$ | $\dfrac{-V_{DC}\left(\frac{R_c^4}{R_\epsilon^4}-1\right)+V_0\left(\frac{R_c^4}{R_\epsilon^4}\right)}{R_\epsilon}$ |

Table 3: Step-by-step solution for forward euler discretization of circuit in Figure 1. Note the diverging oscillations with successive powers of $(R_c/R_\epsilon)$, demonstrating numerical instability.

## Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

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### A.2.1 Simulation Parameters

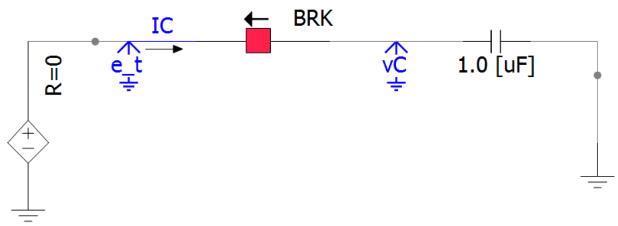


Figure 2: PSCAD schematic for capacitor circuit simulation

I implemented all three discretization methods in MATLAB with the following circuit parameters: $V_{DC} = 10$ V, $V_0 = 0$ V, $C = 1\,\mu$F, and $h = 100\,\mu$, $T = 1ms$. I also simulated the Trapezoidal method in PSCAD as shown in Figure 2 with chatter attenuation disabled to allow direct comparison with the analytical solution.

### A.2.2 Method Comparison and Selection

**Forward Euler: Rejected**

The Forward Euler method exhibits catastrophic numerical instability, as demonstrated in Figure 3. The current diverges exponentially with an amplification factor of $(R_c/R_\epsilon)^n \approx 10^{3n}$ at step $n$ (Note that I've taken $R_\epsilon = R_c/1000$), reaching magnitudes exceeding $10^{17}$ A within microseconds. This non-physical behavior renders the method completely unsuitable for this circuit. Due to its extreme divergence, Forward Euler is shown separately and not included in the comparison plot with the other two methods.

**Backward Euler vs. Trapezoidal**

Figure 4 compares the Backward Euler and Trapezoidal methods against the PSCAD reference solution.

Key observations:

- **Backward Euler**: Provides maximum numerical damping, driving $i(t) \to 0$ after the initial transient. This represents the *closest approach to the true steady-state solution* (zero current after capacitor charges), effectively suppressing the numerical oscillations entirely.

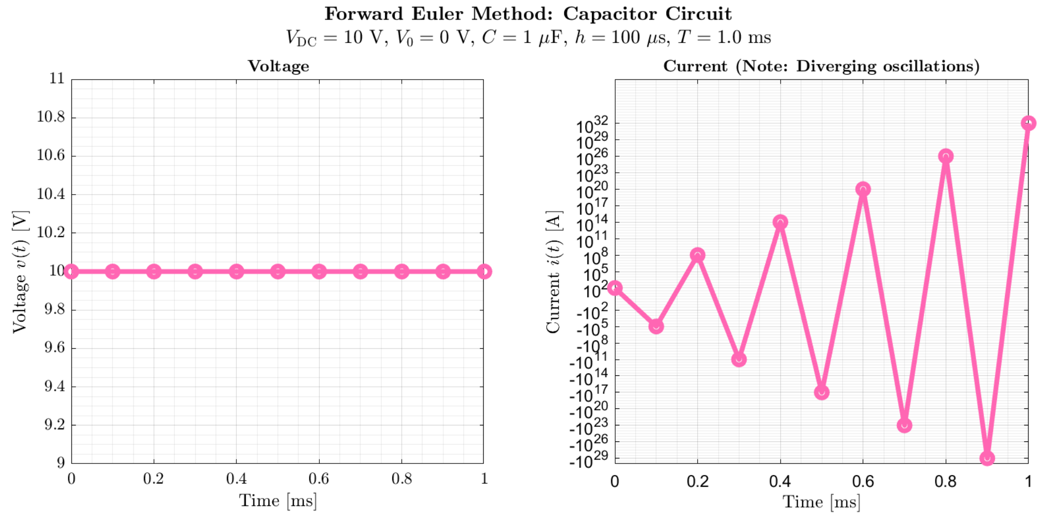


Figure 3: Forward Euler method showing exponential divergence in current magnitude (symmetric logarithmic scale). The oscillations grow with successive powers of $(R_c/R_\epsilon)$, demonstrating fundamental numerical instability.

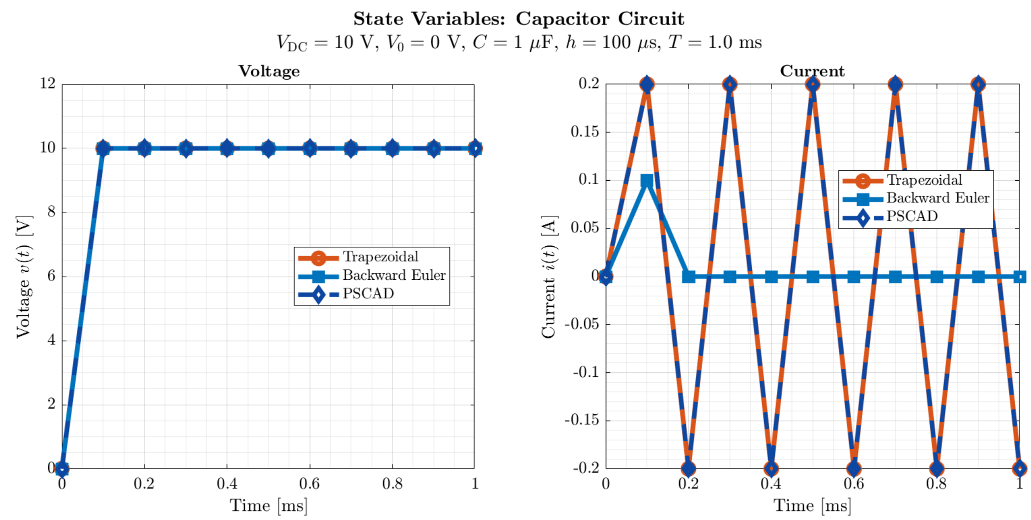


Figure 4: Comparison of Backward Euler, Trapezoidal, and PSCAD solutions. BE provides maximum damping (minimizing current oscillations), while Trapezoidal matches PSCAD exactly and exhibits numerical chatter characteristic of lossless circuit discretization.

- **Trapezoidal**: Exhibits sustained bounded oscillations ("chatter") due to its energy-preserving nature. My MATLAB implementation and PSCAD solution (with chatter attenuation disabled) are *identical*, confirming correct implementation. The chatter is a numerical artifact of discretizing a lossless system.

### A.2.3 Impact of Step Size $h$

The discretization resistance $R_C$ scales linearly with step size $h$, directly coupling time-step selection to numerical behavior. Examining the step-by-step solutions in Tables 1 to 3, the impact of increasing $h$ is summarized in Table 4:

| **Method** | $R_C$ | **Chatter** | **Remark** |
|---|---|---|---|
| **BE** | $h/C$ | **None** | $i(t) = 0$ (damped) as $t \geq 2h$ |
| **Trapz** | $h/(2C)$ | $\downarrow$ | Bounded oscillations persist |
| **FE** | $h/C$ | Unstable | Amplification: $(R_c/R_\epsilon)^n$ |

Table 4: Impact of increasing $h$ on discretization behavior (chatter magnitude $\downarrow$ = desirable).

### A.2.4 Preferred Method

**Backward Euler** is preferred for this specific problem because it most closely represents the true steady-state behavior (zero current after capacitor fully charges), effectively eliminating the non-physical chatter artifacts. While in general Trapezoidal method offers higher accuracy (second-order vs. first-order), in this instance damping is prioritized over accuracy for the conclusion.

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## Section B (Basic Circuit Numerical Simulation): Question 1

The parameters of the circuit below are: $R = 10\Omega$, $L = 20\text{mH}$, $e(t) = 10\text{V}$ dc, and the switch is closed at $t = 0$. The circuit below is to be solved from $t = 0$ to $t = 10\text{ms}$ using: a) PSCAD software, b) your own MATLAB code based on the nodal analysis method.

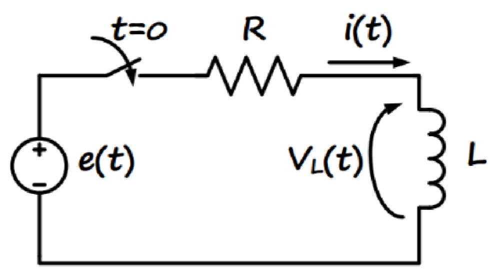


Figure 5: RL circuit with DC voltage source and switch

1) Obtain the solution for the inductor current and voltage using PSCAD with time-step sizes $\Delta t = 0.1\text{ms}$ and $\Delta t = 0.8\text{ms}$.

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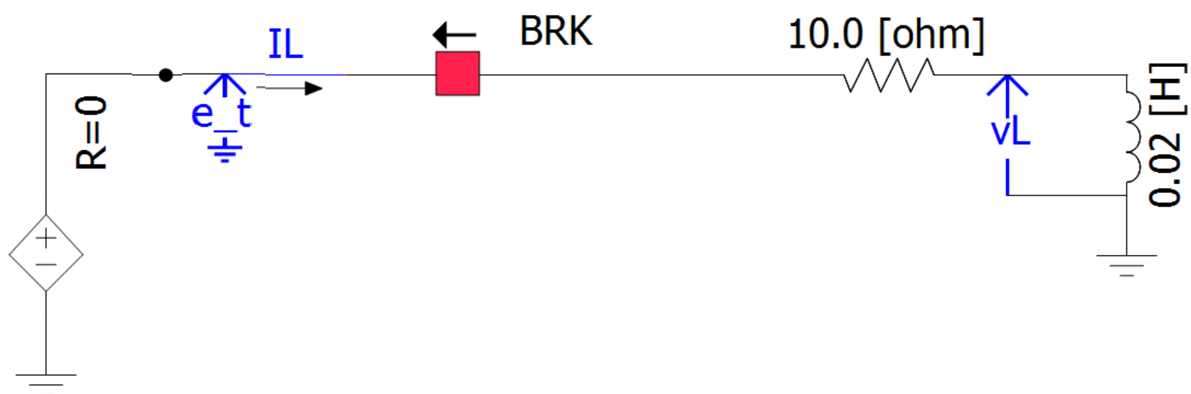


Figure 6: PSCAD schematic for RL circuit simulation

As shown in Figure 6, I implemented the RL circuit in PSCAD with the specified parameters and simulated it using two different time-step sizes: $\Delta t = 0.1$ ms and $\Delta t = 0.8$ ms. The results for the inductor current and voltage waveforms are presented in Figure 7.

**Observations:**

**Inductor Voltage:** The inductor voltage $v_L(t)$ exhibits the expected exponential decay from an initial value of approximately 10 V (at $t = 0^+$) to zero as the circuit reaches steady state. The voltage decays with time constant $\tau = L/R = 2$ ms. Both time-step sizes capture this behavior accurately, with the finer time step ($\Delta t = 0.1$ ms) providing smoother resolution of the transient. The coarser time step ($\Delta t = 0.8$ ms) still captures the overall decay trend but with fewer sample points.

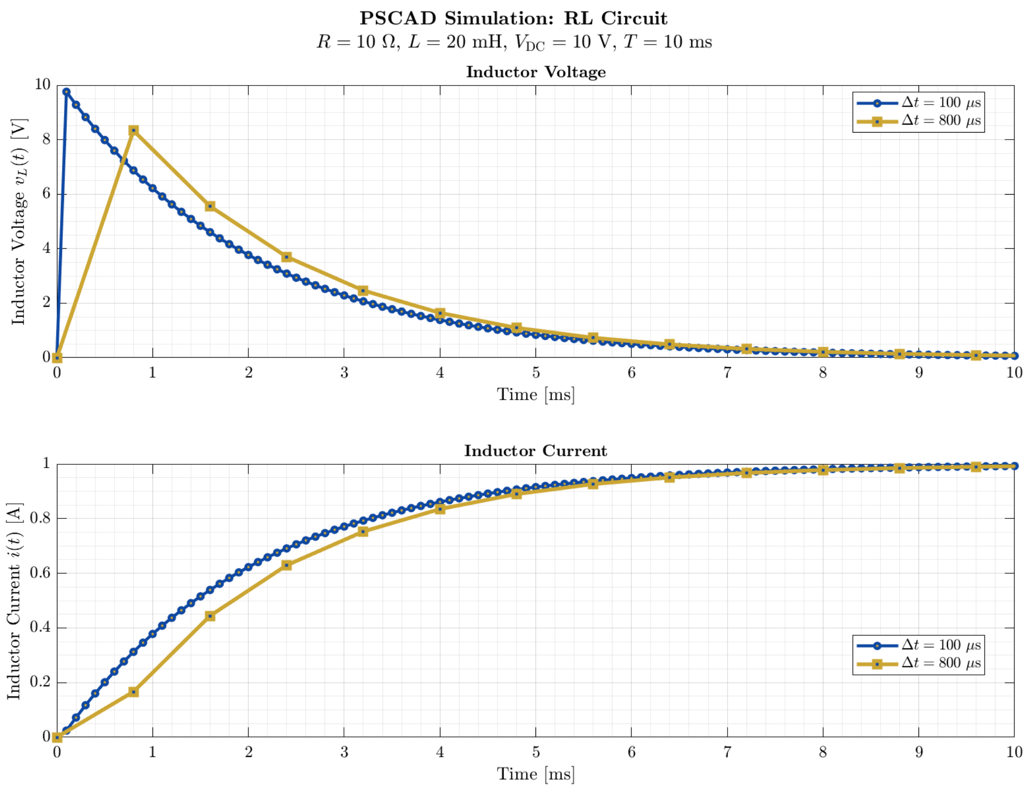


Figure 7: PSCAD simulation results of Figure 6 for inductor voltage and current with two time-step sizes

**Inductor Current:** The inductor current $i(t)$ rises exponentially from zero to its steady-state value of $I_{\text{ss}} = V_{\text{DC}}/R = 1$ A, following the complementary behavior to the voltage. Both time-step sizes converge to the same steady-state value, demonstrating that PSCAD's trapezoidal integration method is stable for this problem. The finer time step provides better resolution of the initial transient behavior.

**Step Size Impact:** The coarser time step ($\Delta t = 0.8$ ms) is close to half the time constant ($\tau/2.5 = 0.8$ ms), yet still produces acceptable results without numerical instability. This demonstrates the robustness of PSCAD's implicit integration scheme for this first-order RL circuit.

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# Section B : Question 2

Discretize the circuit in Figure 5 and write a MATLAB code to solve the circuit step-by-step using the two discretization rules: a) Backward Euler, b) Trapezoidal, each with time-step sizes $\Delta t = 0.1\,\text{ms}$ and $\Delta t = 0.8\,\text{ms}$.

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## B.2.1 Inductor Discretization Formulas

### Trapezoidal Rule

For an inductor with $v(t) = L\frac{di(t)}{dt}$, the trapezoidal discretization (derivation in Appendix IV) yields:

**Trapezoidal: Inductor Companion Model**

$$\boxed{v(t) = R_L i(t) - e_h} \tag{17}$$

$$\boxed{i(t) = \frac{v(t)}{R_L} + i_h} \tag{18}$$

where:

$$R_L = \frac{2L}{h} \tag{19}$$

$$e_h = R_L i(t-h) + v(t-h) \tag{20}$$

$$i_h = \frac{e_h}{R_L} = i(t-h) + \frac{v(t-h)}{R_L} \tag{21}$$

### Backward Euler

For the same inductor, the Backward Euler discretization (derivation in Appendix V) yields:

**Backward Euler: Inductor Companion Model**

$$\boxed{v(t) = R_L i(t) - e_h} \tag{22}$$

$$\boxed{i(t) = \frac{v(t)}{R_L} + i_h} \tag{23}$$

where:

$$R_L = \frac{L}{h} \tag{24}$$

$$e_h = R_L i(t-h) \tag{25}$$

$$i_h = i(t-h) = \frac{e_h}{R_L} \tag{26}$$

### B.2.2 State Variable Trajectory Generating Equations

**Note on Circuit and Equations:** The circuit topology in Figure 5 is identical for both Backward Euler and Trapezoidal discretization methods - only the companion model parameters ($R_L$, $e_h$, $i_h$) differ between the two methods. The state variable trajectory generating (SVTG) equations derived from KVL are also identical for both methods, as shown below. Unlike Question A.1 where symbolic step-by-step tables were feasible, the expressions here become algebraically complex due to the resistor-inductor interaction. Therefore, I present only the generating equations and proceed directly to numerical MATLAB trajectory generation.

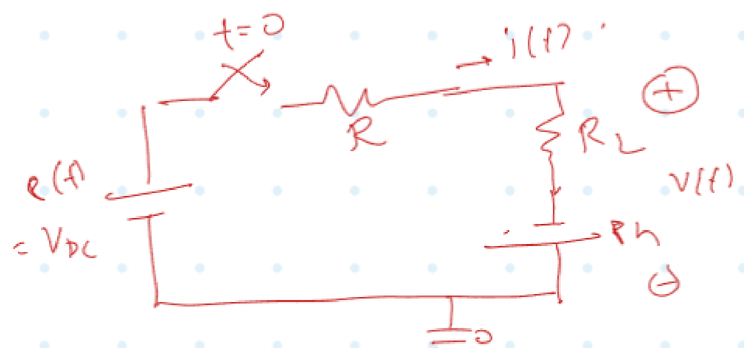


Figure 8: Circuit analysis with labeled variables showing companion model components for discretization

The circuit consists of a resistor $R$, an inductor $L$, and a DC voltage source $e(t)$. Using the companion model formulas above, the state variables at time $t$ are calculated using the following causality:

**Important Note on Causality:** At each time step $t$, assuming $e_h$ and $i_h$ are already computed from the previous step, we calculate in this order:

1. Calculate $i(t)$ first (using KVL and companion model)
2. Then calculate $v_L(t)$ (using the companion model voltage equation)

**Trapezoidal Rule - Table Generating Equations**

For the RL circuit with KVL: $V_{DC} - i(t) \cdot R - v_L(t) = 0$

Substituting the companion model $v_L(t) = R_L i(t) - e_h$ into KVL:

$$V_{DC} - i(t) \cdot R - (R_L i(t) - e_h) = 0 \tag{27}$$

$$V_{DC} + e_h = i(t)(R + R_L) \tag{28}$$

**State variable calculation sequence:**

**Trapezoidal: State Variables at Time $t$ for Figure 5**

$$\boxed{i(t) = \frac{V_{DC} + e_h}{R + R_L}} \quad \text{(calculate current first)} \tag{29}$$

$$\boxed{v_L(t) = R_L i(t) - e_h} \quad \text{(then calculate voltage)} \tag{30}$$

**History terms for next step:**

$$e_h = R_L i(t) + v_L(t) \quad \text{(for use at } t + h\text{)} \tag{31}$$

$$i_h = i(t) + \frac{v_L(t)}{R_L} \quad \text{(for use at } t + h\text{)} \tag{32}$$

**Initial conditions at** $t = 0^-$:

$$v_L(0^-) = 0, \quad i(0^-) = 0 \tag{33}$$

**Backward Euler - Table Generating Equations**

**Important:** The state variable calculation equations are *identical* to Trapezoidal (shown above). The only difference lies in the history term updates, which depend on the method-specific companion model parameters.

**State variable calculation sequence:**

**Backward Euler: State Variables at Time $t$ for Figure 5**

$$\boxed{i(t) = \frac{V_{DC} + e_h}{R + R_L}} \quad \text{(calculate current first)} \tag{34}$$

$$\boxed{v_L(t) = R_L i(t) - e_h} \quad \text{(then calculate voltage)} \tag{35}$$

**History terms for next step:**

$$e_h = R_L i(t) \quad \text{(for use at } t+h\text{)} \tag{36}$$

$$i_h = i(t) = \frac{e_h}{R_L} \quad \text{(for use at } t+h\text{)} \tag{37}$$

**Initial conditions at** $t = 0^-$:

$$v_L(0^-) = 0, \quad i(0^-) = 0 \tag{38}$$

### B.2.3 MATLAB Implementation

Figure 9 represents the MATLAB implementation in Section B.2.

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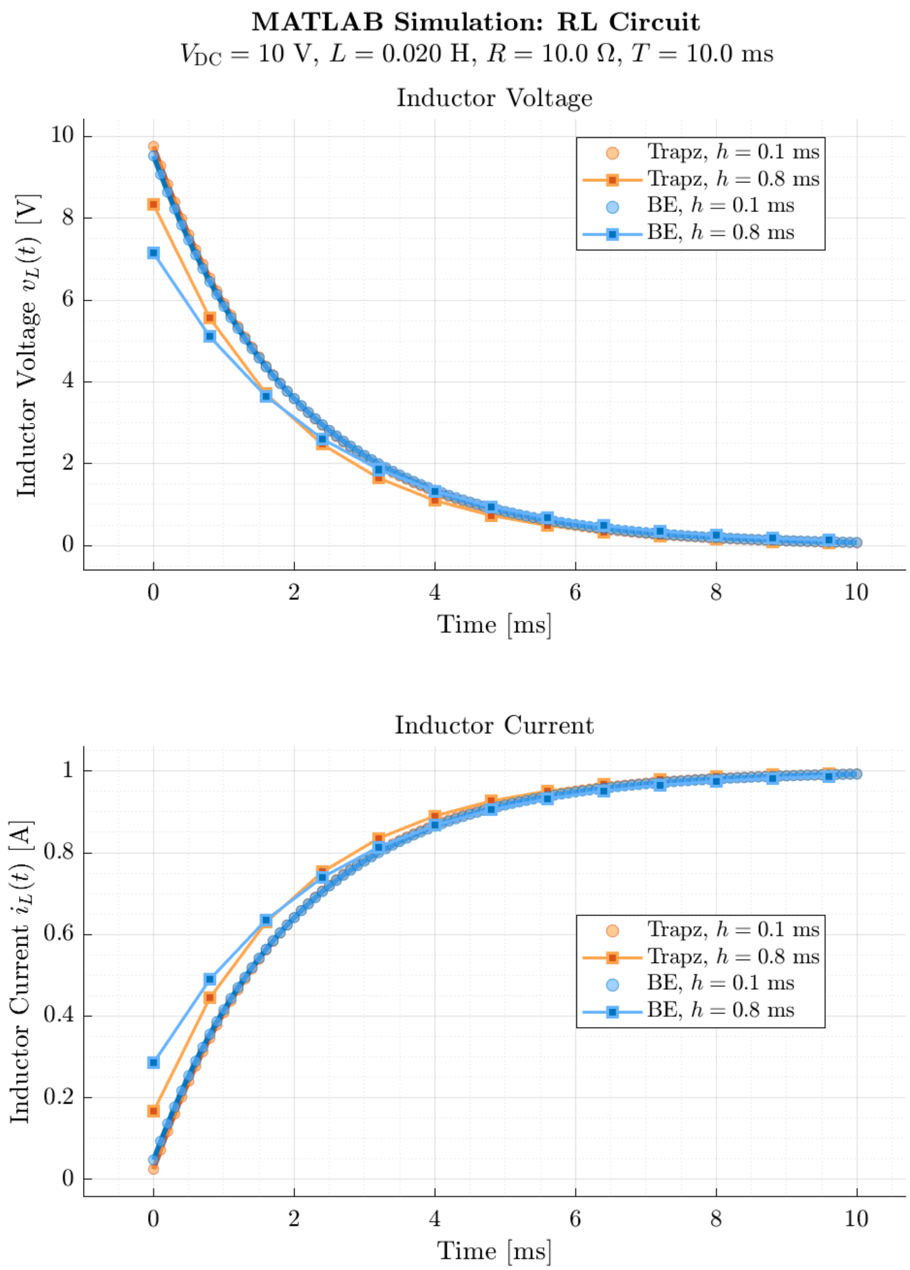


Figure 9: MATLAB implementation of Figure 5 for Trapezoidal and Backward Euler methods at $h = 0.1$ ms and $h = 0.8$ ms. Markers indicate all computed time steps.

## Section B: Question 3

3) Show the results for $i(t)$ obtained in the six cases (PSCAD, your Backward Euler code, and your Trapezoidal code; each with two different step sizes) superimposed in a (big enough) Plot. Show the same for $v_L(t)$ in a separate plot. Comment on the accuracy of the methods, impact of the time-step size, and how the PSCAD results compare with your Trapezoidal code.

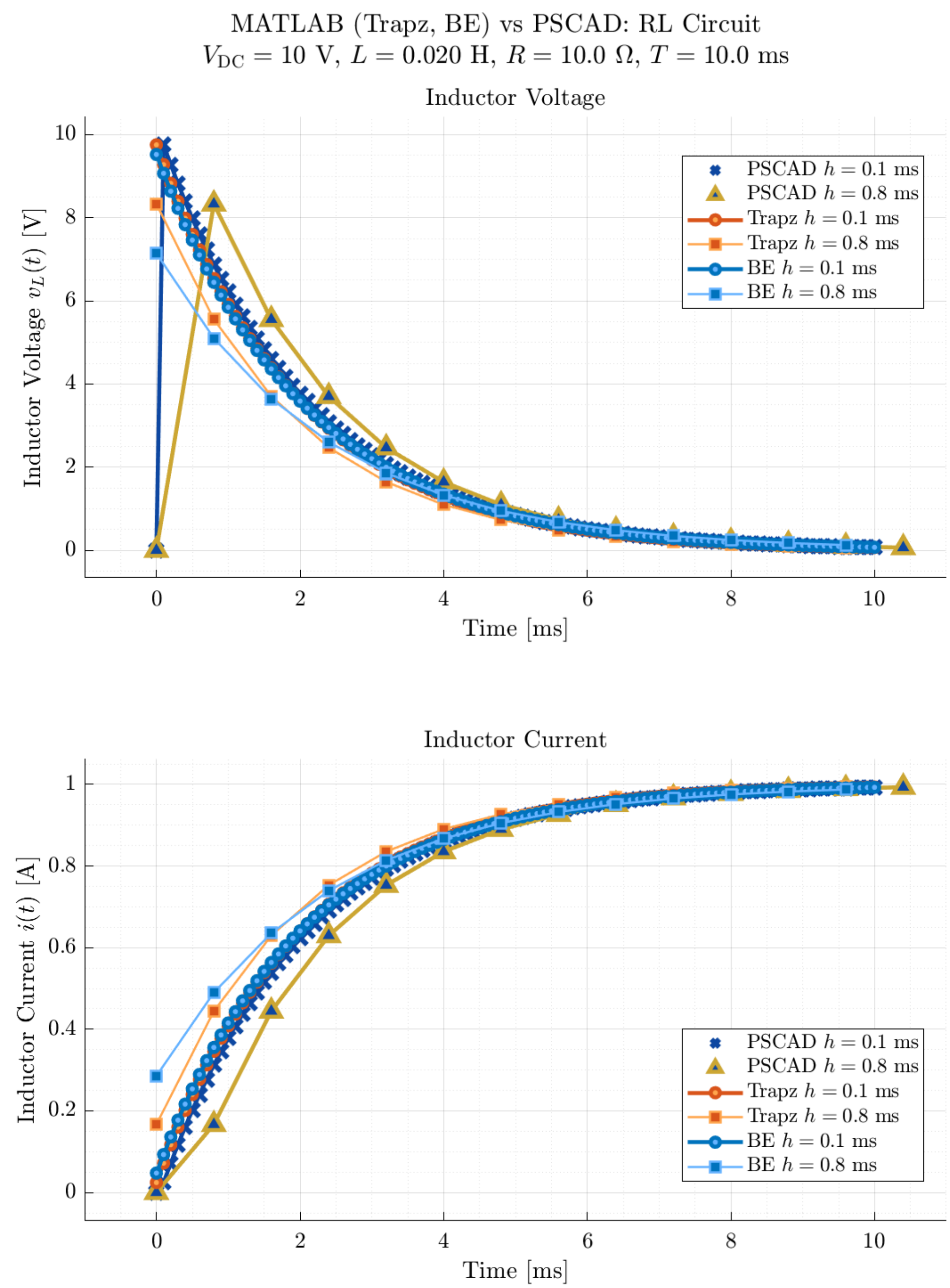


Figure 10: Comparison of inductor current $i(t)$ and voltage $v_L(t)$ for the RL circuit using PSCAD (two time steps), MATLAB Trapezoidal, and Backward Euler (two time steps each). All six results are superimposed for direct comparison.

**Discussion:**

*Accuracy:* For $h = 0.1$ ms, all methods closely match, with Trapezoidal and PSCAD nearly indistinguishable. For $h = 0.8$ ms, the Backward Euler method shows more numerical damping, while Trapezoidal and PSCAD remain close, though some time-step artifacts are visible.

*Time-step Impact:* Larger $h$ increases numerical error, especially for Backward Euler (overdamping). Trapezoidal is more accurate for both $i(t)$ and $v_L(t)$ at larger $h$.

*PSCAD vs Trapezoidal:* The MATLAB Trapezoidal method and PSCAD trends are very similar for both time steps, with the only discrepancy probably caused due to the different state variable initialization by PSCAD - technically PSCAD seems to use $t = 0^-$ whereas I use $t = 0^+$ in my matlab code - causing a significant initial discrepancy which takes a little while to get damped out as it keeps manifesting as a history term in PSCAD solution. But in general I'd consider them to be in very good agreement.

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# Section C (Circuit Interruption): Question 1

The case below (figure on the left) represents a typical short-circuit fault in transmission lines and the breaker operation for line interruption. The equivalent circuit is shown in the figure on the right.

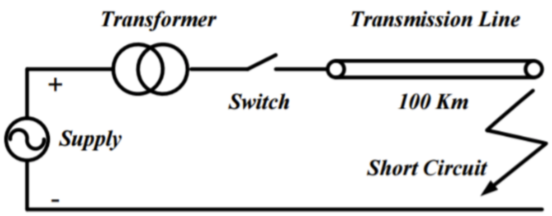


Figure 11: Transmission line with short-circuit fault

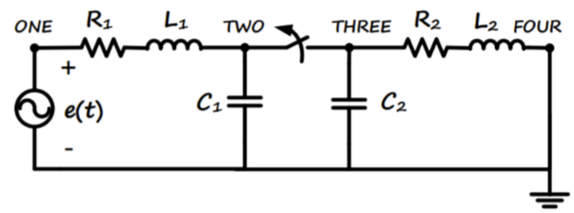


Figure 12: Equivalent circuit representation

On the left-hand side of the switch/breaker, we have the source voltage and a resistance and inductance representing the supply system combined with the transformer. The capacitor to the left of the breaker represents the combined substation capacitance due to the substation busbars. On the right-hand side of the breaker, we have the transmission line parameters: resistance, inductance, and capacitance represented as simple lumped parameters, where the right side is short-circuited due to the fault, assuming that the fault has already been applied. We have assumed a single-phase circuit here for simplicity. The parameters of the circuit below are: $e(t) = 230 \times \cos(\omega t)$ kV, $\omega = 377$ rad/s, $R_1 = 3\Omega$, $L_1 = 350$ mH, $R_2 = 50\Omega$, $L_2 = 100$ mH, $C_1 = 10$ nF, $C_2 = 600$ nF. Assume that the system is operating with zero initial conditions and the breaker is initially closed. At $t = 8$ ms, the breaker is commanded to open the line, but it will not actually open until the current passes through zero. The study period is from $t = 0$ to $t = 20$ ms.

1) Choose an appropriate simulation time-step size $\Delta t$ (assuming the Trapezoidal integration rule) according to the time constants in the circuit. For this, evaluate the approximate resonant frequencies analytically before and after the breaker operation.

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### C.1.1 Solution Approach

To determine an appropriate time-step for the trapezoidal discretization, we need to analyze the resonant frequencies of the circuit in three different states:

1. **Closed state**: Breaker closed (full circuit)
2. **Open state (Left)**: Left portion of the circuit after breaker opens
3. **Open state (Right)**: Right portion of the circuit after breaker opens

The circuit configurations are shown in Figures 13 and 14.

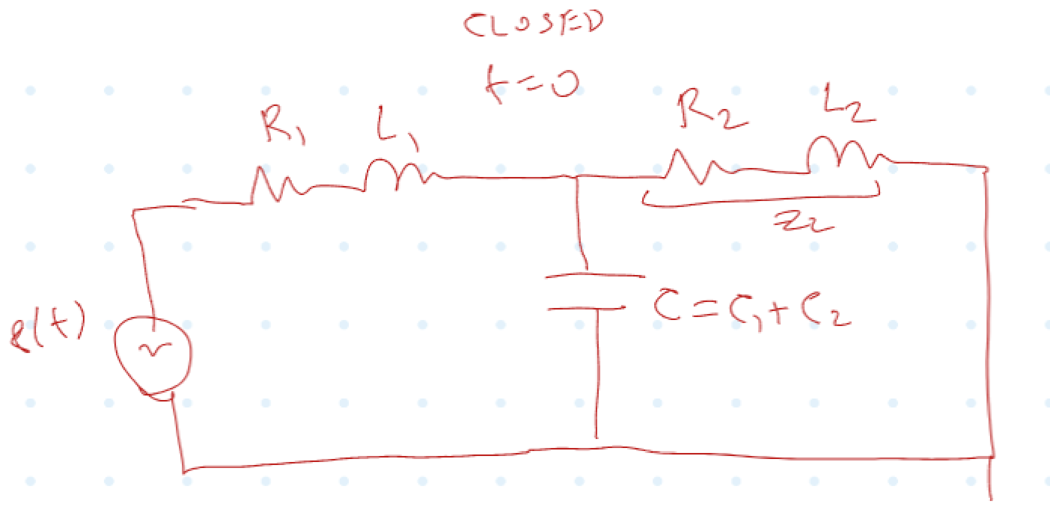


Figure 13: Closed circuit configuration (breaker closed)

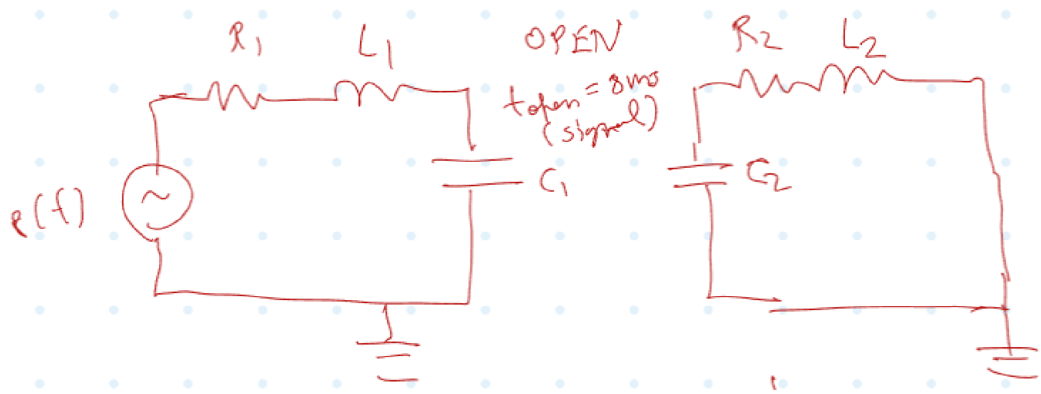


Figure 14: Open circuit configuration (breaker open, showing left and right portions)

The criterion for time-step selection is based on the maximum frequency component in the circuit to ensure numerical accuracy with the trapezoidal rule.

### C.1.2 Resonant Frequency Calculations

For an LC circuit, the resonant frequency is given by:

$$f_{\text{res}} = \frac{1}{2\pi\sqrt{LC}} \tag{39}$$

#### C.1.2.1 Closed State (Breaker Closed)

When the breaker is closed, the inductances are in parallel and capacitances are in series:

$$L_{\text{eq}} = \frac{L_1 L_2}{L_1 + L_2} = \frac{(0.350)(0.100)}{0.350 + 0.100} = 0.0778 \text{ H} \tag{40}$$

$$C_{\text{eq}} = C_1 + C_2 = 10 \times 10^{-9} + 600 \times 10^{-9} = 6.10 \times 10^{-7} \text{ F} \tag{41}$$

The resonant frequency is:

$$f_{\text{res,closed}} = \frac{1}{2\pi\sqrt{(0.0778)(6.10 \times 10^{-7})}} = 730.68 \text{ Hz} \tag{42}$$

#### C.1.2.2 Open State - Left Portion

When the breaker is open, the left portion consists of $L_1$ and $C_1$:

$$f_{\text{res,left}} = \frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{1}{2\pi\sqrt{(0.350)(10 \times 10^{-9})}} = 2690.21 \text{ Hz} \tag{43}$$

#### C.1.2.3 Open State - Right Portion

The right portion consists of $L_2$ and $C_2$:

$$f_{\text{res,right}} = \frac{1}{2\pi\sqrt{L_2 C_2}} = \frac{1}{2\pi\sqrt{(0.100)(600 \times 10^{-9})}} = 649.75 \text{ Hz} \tag{44}$$

### C.1.3 Time-Step Selection Criterion

For the trapezoidal rule with 3% error tolerance, the recommended time-step is:

$$h = \frac{1}{10 f_{\max}} \tag{45}$$

where $f_{\max} = f_{\text{res}}$ for each circuit state.

### C.1.4 Recommended Time-Steps

Computing the time-step for each state:

$$h_{\text{closed}} = \frac{1}{10 \times 730.68} = 1.369 \times 10^{-4} \text{ s} = 136.86\,\mu\text{s} \tag{46}$$

$$h_{\text{left}} = \frac{1}{10 \times 2690.21} = 3.717 \times 10^{-5} \text{ s} = 37.17\,\mu\text{s} \tag{47}$$

$$h_{\text{right}} = \frac{1}{10 \times 649.75} = 1.539 \times 10^{-4} \text{ s} = 153.91\,\mu\text{s} \tag{48}$$

### C.1.5 Overall Recommendation

To ensure accurate simulation throughout all circuit states (before and after breaker operation), we must use the most conservative (minimum) time-step:

**Simulation Time-Step Recommendation**

$$\boxed{h = 37.17\,\mu\text{s}}$$

This corresponds to the open state (left portion), which has the highest resonant frequency.

## Section C: Question 2

2) Write your own MATLAB code to solve the system and also obtain the solution using PSCAD.

## Section C: Question 3

3) Show the results for voltages at nodes $Two$ and $Three$ and the fault current (i.e., from node $Four$ to $ground$) in three separate (big enough) plots. In each plot, superimpose the results from your code and also from PSCAD, both obtained using the same simulation time-step size obtained in part 1) for comparison. Comment on the accuracy of your results compared to those of PSCAD.

# I Appendix: Derivation of Trapezoidal Discretization for Capacitor

To derive the equations $v_t$ and $i_t$, the following steps are involved:

- Start with the governing equation for a capacitor:

$$i(t) = C\frac{dv(t)}{dt} \tag{49}$$

- Apply the trapezoidal rule to discretize the relationship between voltage and current:

$$v(t) - v(t - \Delta t) = \frac{\Delta t}{2C}\left[i(t) + i(t - \Delta t)\right] \tag{50}$$

- Define the equivalent resistance $R_C$ and history terms:

$$R_C = \frac{\Delta t}{2C} \tag{51}$$

- Derive the Thevenin (series) and Norton (parallel) companion models.

The final equations for the capacitor are:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{52}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{53}$$

where:

- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$

# II Appendix: Derivation of Backward-Euler Discretization for Capacitor

Starting from the capacitor constitutive relation

$$i(t) = C\frac{dv(t)}{dt}, \tag{54}$$

the Backward-Euler (BE) discretization at time $t$ approximates the derivative by

$$\frac{dv(t)}{dt} \approx \frac{v(t) - v(t - \Delta t)}{\Delta t}. \tag{55}$$

Substituting and rearranging yields

$$i(t) \approx C\frac{v(t) - v(t - \Delta t)}{\Delta t}. \tag{56}$$

Writing the BE companion in a Thévenin/Norton form gives an equivalent resistance and history terms. The final boxed companion equations used in the main text (capacitor, BE) are

$$\boxed{v(t) = e_h + R_C\, i(t)} \tag{57}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{58}$$

with the BE companion parameters (for the capacitor):

$$R_C = \frac{\Delta t}{C}, \tag{59}$$

$$e_h = v(t - \Delta t), \tag{60}$$

$$i_h = \frac{v(t - \Delta t)}{R_C}. \tag{61}$$

These expressions are used when implementing the Backward-Euler discretization in the numerical examples and the BE table in the main text.

# III Appendix: Derivation of Forward-Euler Discretization for Capacitor

Starting from the capacitor constitutive relation

$$i(t) = C\frac{dv(t)}{dt}, \tag{62}$$

the Forward-Euler (FE) discretization at time $t$ approximates the derivative using the previous time step:

$$\frac{dv(t)}{dt} \approx \frac{v(t) - v(t-h)}{h}. \tag{63}$$

Substituting and rearranging yields

$$i(t) \approx C\frac{v(t) - v(t-h)}{h}. \tag{64}$$

To cast this in a companion model form similar to BE and trapezoidal methods, I introduce an artificial small resistance $R_\epsilon \ll h/C$ (as justified in the main text to avoid division by zero). The final boxed companion equations used in the main text (capacitor, FE) are

$$\boxed{v(t) = e_h(t)} \tag{65}$$

$$\boxed{i(t) = \frac{v(t) - e_h(t)}{R_\epsilon}} \tag{66}$$

with the FE companion parameters (for the capacitor):

$$R_c = \frac{h}{C} \quad , \tag{67}$$

$$e_h(t) = v(t-h) + R_c\, i(t-h), \tag{68}$$

$$i_h(t) = \frac{e_h(t)}{R_c}. \tag{69}$$

These expressions are used when implementing the Forward-Euler discretization in the numerical examples and the FE table in the main text.

# IV Appendix: Derivation of Trapezoidal Discretization for Inductor

Starting from the inductor constitutive relation

$$v(t) = L\frac{di(t)}{dt}, \tag{70}$$

we integrate from $t - h$ to $t$ to obtain:

$$\int_{t-h}^{t} v(\tau)\, d\tau = L\left[i(t) - i(t-h)\right]. \tag{71}$$

The trapezoidal rule approximates the integral as:

$$\int_{t-h}^{t} v(\tau)\, d\tau \approx \frac{h}{2}\left[v(t-h) + v(t)\right]. \tag{72}$$

Substituting and rearranging:

$$i(t) - i(t-h) = \frac{h}{2L}\left[v(t-h) + v(t)\right]. \tag{73}$$

Multiply both sides by $\frac{2L}{h}$ and rearrange:

$$\frac{2L}{h}\left(i(t) - i(t-h)\right) = v(t-h) + v(t). \tag{74}$$

Define the equivalent resistance $R_L = \frac{2L}{h}$:

$$R_L\left(i(t) - i(t-h)\right) = v(t-h) + v(t). \tag{75}$$

Solving for $v(t)$:

$$v(t) = R_L i(t) - \underbrace{\left[R_L i(t-h) + v(t-h)\right]}_{e_h}. \tag{76}$$

The companion model parameters are:

$$R_L = \frac{2L}{h} \tag{77}$$
$$e_h = R_L i(t-h) + v(t-h) \tag{78}$$
$$i_h = \frac{e_h}{R_L} = i(t-h) + \frac{v(t-h)}{R_L} \tag{79}$$

The final boxed equations are:

$$\boxed{v(t) = R_L i(t) - e_h} \tag{80}$$

$$\boxed{i(t) = \frac{v(t)}{R_L} + i_h} \tag{81}$$

# V Appendix: Derivation of Backward Euler Discretization for Inductor

Starting from the inductor constitutive relation

$$v(t) = L\frac{di(t)}{dt}, \tag{82}$$

the Backward Euler discretization approximates the derivative at time $t$ as:

$$\frac{di(t)}{dt} \approx \frac{i(t) - i(t-h)}{h}. \tag{83}$$

Substituting:

$$v(t) \approx L\frac{i(t) - i(t-h)}{h}. \tag{84}$$

Rearranging:

$$v(t) = \frac{L}{h}i(t) - \frac{L}{h}i(t-h). \tag{85}$$

Define the equivalent resistance $R_L = \frac{L}{h}$:

$$v(t) = R_L i(t) - \underbrace{R_L i(t-h)}_{e_h}. \tag{86}$$

The companion model parameters are:

$$R_L = \frac{L}{h} \tag{87}$$

$$e_h = R_L i(t-h) \tag{88}$$

$$i_h = i(t-h) = \frac{e_h}{R_L} \tag{89}$$

The final boxed equations are:

$$\boxed{v(t) = R_L i(t) - e_h} \tag{90}$$

$$\boxed{i(t) = \frac{v(t)}{R_L} + i_h} \tag{91}$$